

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
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I want to write a doctorate on how the design of renewable support schemes changes what actually gets built, and I would bring identification to it. Energy market design is argued mostly in theory, while Europe has spent a decade running the experiments: member states adopted auctions at different dates, under different rules, and revised them on their own timetables. That is variation an applied microeconomist can use. It is also why I am writing to this project rather than to a general energy post. The question underneath both of the directions I would want to take up is the one my own work has circled, which is that the support an instrument appears to offer and the incentive it creates are different quantities, and the rules decide the distance between them.

My training is in economics at the University of Cologne. The energy specialisation there runs across two modules. *Growth, Energy, Climate Change* covers the physical foundations of energy systems, their economic foundations, and European regulation and market intervention. *Energy Markets and Regulation*, with Jun.-Prof. Dr. Oliver Ruhnau, takes those further into grids and interconnectors and the European electricity market, and I finished it with 1.3 on a scale where 1.0 is the maximum. My quantitative preparation is the strongest part of my record: mathematics for economists 1.0, econometrics 1.3, and at bachelor level statistics 1.0, mathematics 1.0 and applied econometrics 1.3. My master's thesis was graded 1.0. It estimates the labour-market effects of a national scholarship programme in Brazil on a panel I built from administrative records, roughly five million linked employer-employee observations across 558 microregions from 2005 to 2019, in R through BigQuery. Using Callaway and Sant'Anna (2021) with doubly-robust inference, the programme raises wages by 3.04 per cent in low-exposure regions, distinguishable from zero at the 10 per cent level, on a dose-response that is non-monotonic in exposure.

The two directions I would want to work on are de-risking green investment and multi-technology auction design. My final seminar paper is already inside the first of them. It defines the effective EU ETS carbon cost as the allowance price attenuated by the sector-year free-allocation share, and shows that the attenuation is not lump-sum, because activity thresholds, closure provisions and partial-cessation rules tie allocation to continued production. A public intervention that weakens a cost incentive while the headline price is unchanged is precisely the mechanism your first direction asks about, and the same logic applies to guarantees, contracts for difference and public insurance. The second direction has the cleaner empirical footing. Since the 2016 state aid framework, member states have arrived at technology-neutral, technology-specific and mixed designs at different dates, and Germany's joint wind and solar auction of April 2018 handed the entire 200 MW to solar at bids above the technology-specific benchmark. Portfolio composition and rents are observable, which makes this measurable and not only modellable.

Three people at CORE make this the place to do it. Your work with Yueting Yu on discriminatory against uniform-price auctions asks how short-run bidding maps into long-run investment, which is the rent question in the second direction. Quentin L  t   works on market design, capacity investment and risk management, which is the first. Enerpol, with Jean Hindriks and Fran  ois Maniquet, studies who carries the cost of the transition, and distributional estimation across regions and groups is what my thesis does. In ten years I want to be producing the evidence European energy policy is made on, from a group of this kind or from inside a European institution, and I would rather reach the second having done the research than having only read it. This position is the first step for both, and I would be glad to discuss it with you.

Available from October 2026. French B1, German and Tunisian Arabic native, English IELTS Academic 8.0. EU citizen. Referees: Prof. Dr. Pia Pinger and Jun.-Prof. Dr. Oliver Ruhnau, contact details on the CV.

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